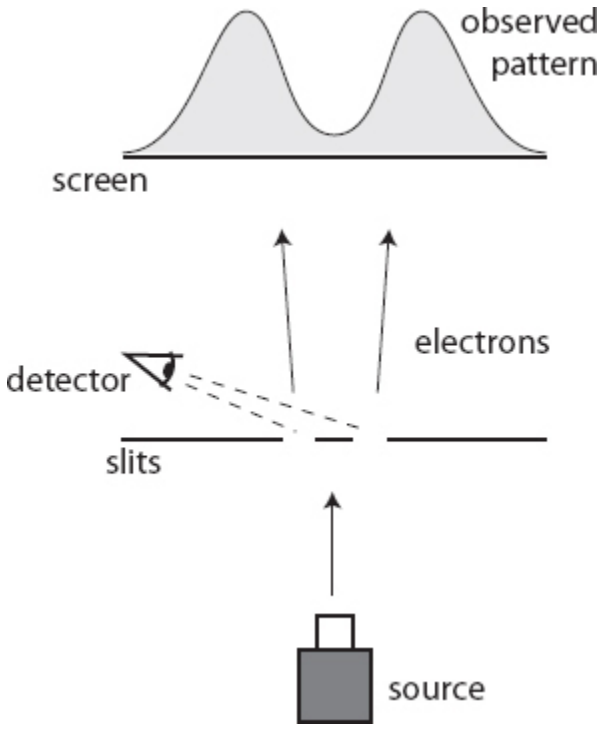


The double-slit experiment makes it difficult to cling to the belief that the electron is just a single classical point, and the wave function simply represents our ignorance about where that point is. Ignorance doesn't cause interference patterns. There is something real about the wave function.

Wave functions may be real, but they're undeniably abstract, and once we start considering more than one particle at a time they become hard to visualize. As we move forward with increasingly subtle examples of quantum phenomena in action, it will be very helpful to have a simple, readily graspable example we can refer to over and over. The *spin* of a particle—a degree of freedom in addition to its position or momentum—is just what we're looking for. We have to think a bit about what spin means within quantum mechanics, but once we do, it will make our lives much easier.

The notion of spin itself isn't hard to grasp: it's just rotation around an axis, as the Earth does every day or a pirouetting ballet dancer does on their tiptoes. But just like the energies of an electron orbiting an atomic nucleus,